

Assignment 2(COMPUTER SYSTEM ARCHITECTURE)

Instructions:

All questions are compulsory

Write your name and university roll no on each page.

Write the solution in handwritten form and upload it on GLA portal.

Qus 1 . Compare and contrast hardwired vs microprogrammed control unit.

Qus 2. Compare the various addressing modes and analyze their advantages and disadvantages.

Qus 3. (i) Draw block diagram of SRAM and how it differs from DRAM.

(ii) What is instruction code and What is the difference between a direct and an indirect address in instruction?

Qus 4 Explain the concept of asynchronous data transfer and Describe how it differs from synchronous data transfer.

Qus 5 Differentiate between RISC and CISC.

Qus 6 . Write a program to evaluate the arithmetic statement:

$$X = A - B + C * (D * E - F) / (G + H * K)$$

a. Using a general register computer with three address instructions.

b. Using a general register computer with two address instructions.

c. Using an accumulator type computer with one address instructions.

d. Using a stack organized computer with zero-address operation instructions.

Qus 7. Explain DMA mode of transfer in detail. Why are the read and write control lines in a DMA controller bidirectional? Under what condition and for what purpose are they used as inputs? Under what condition and for what purpose are they used as outputs?

Qus 8. What are the various phases in instruction cycle? Explain with the flow chart of Instruction execution.

Qus 9. (i) Explain pipelining and its advantages.

(ii) Explain Data Hazard How do we overcome it? Discuss its side effects.

Qus 10. (i) Explain interface in I/O organization. Why is it required?

(ii) A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 eight bit words.

How many bits are required for addressing the main memory?

How many bits are needed to represent the TAG, SET and WORD fields?